MODULE-I

Introduction to Algorithm Analysis

Algorithm and its Properties

Algorithm

1. **Step by step procedure** for solving any problem is known as algorithm.
2. An algorithm is a **finite set of instructions** that, if followed, accomplishes a particular task.
3. An algorithm is a **sequence of computational steps** that transform the input into a valuable or required output.
4. Any **special method of solving a certain kind of problem** is known as algorithm.

All Algorithms must satisfy the following criteria

1) **Input:** There are more quantities that are externally supplied.(zero or more)

2) **Output:** At least one quantity is produced.

3) **Definiteness:** Each instruction of the algorithm should be clear and unambiguous.

4) **Finiteness:** The process should be terminated after a finite number of steps.

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5) **Effectiveness:** Every instruction must be basic enough to be carried out theoretically or by using paper and pencil.

Properties of Algorithm

It is necessary to have following properties associated with an algorithm.

Non Ambiguity

- Each step in an algorithm should be non-ambiguous.
- That means each instruction should be clear and precise.
- The instruction in any algorithm should not denote any conflicting meaning. This property also indicates the effectiveness of algorithm.

Range of Input

- The range of input should be specified.
- Normally the algorithm is input driven.
- If the range of input is not being specified then algorithm can go in an infinite state.

Multiplicity

- The same algorithm can be represented into several different ways.
- We can write in simple English the sequence of instruction or we can write it in form of pseudo code.
- For solving the same problem we can write several different algorithms.

Speed

- Algorithm should be efficient and should produce the output with fast speed.

Finiteness

- The algorithm should be finite. That means after performing required operations it should be terminate.